

MINI PROJECT - REPORT
on
CARDIOKEY: ANALOG ECG SIGNAL
ACQUISITION AND BIOMETRICS

Submitted by

AARDRA S V – 24101868
AAROMAL A – 24101880
ABDULLA HILAL V K – 24101000
ABHIJITH R S – 24102771

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DIVISION OF ELECTRONICS ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
KOCHI-682022

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**DIVISION OF ELECTRONICS ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
KOCHI-682022**

CERTIFICATE

*Certified that the mini project report entitled “**CARDIOKEY: ANALOG ECG SIGNAL ACQUISITION AND BIOMETRICS**” is a bonafide work of **AARDRA S V (24101868)**, **AAROMAL A (24101880)**, **ABDULLA HILAL V K (24101000)**, and **ABHIJITH R S (24102771)**, towards the partial fulfilment for the award of the degree of B. Tech in Electronics and Communication of Cochin University of Science and Technology, Kochi-682022.*

Project Guide

Ms. Reshmi R. Das

Head of the Division

Dr. Deepa Sankar

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ABSTRACT

Traditional biometrics lack inherent liveness detection, making the Electrocardiogram (ECG)—with its unique P, QRS, and T wave morphology dictated by individual cardiac anatomy—a highly secure alternative. The "CardioKey" project proposes a comprehensive biometric system utilizing an AD8232 ECG sensor and a precision analog front-end to amplify Lead I heart signals while rejecting 60Hz mains hum, which is then digitized at 300 Hz using a 16-bit ADS1115 ADC and an Arduino UNO data pipeline. Through Digital Signal Processing (DSP) involving wavelet drift correction and digital filtering, 150-sample P-QRS-T fragments are segmented per cardiac cycle and compressed into 30-point biometric signatures using Principal Component Analysis (PCA) to be evaluated using Linear Discriminant Analysis (LDA) and Nearest Mean classifiers. Validated on the public ECGID database of 45 subjects using Python, the system achieves up to 96% identification accuracy under resting conditions, and by addressing heart rate variability through normalization and QT scaling, proves the ECG is a viable, universal, and unforgeable biometric trait for practical access control systems.

CONTENTS

LIST OF FIGURES	vi
LIST OF TABLES	viii
ABBREVIATIONS	ix
1 INTRODUCTION	1
2 LITERATURE REVIEW	3
3 PROBLEM IDENTIFICATION AND OBJECTIVE	5
3.1 Objectives	6
4 WHY WE CHOSE THIS PROJECT	7
5 SENSORS AND EQUIPMENT OVERVIEW	8
5.1 Analog Front-End: AD8232 / AD620	8
5.2 High-Resolution Digitization: ADS1115	9
5.3 Microcontroller Unit (MCU)	9
6 ANALOG– CIRCUIT DESIGN AND TESTING	11
6.1 The Differential Input & Output Amplifier Stages	11
6.2 The Right Leg Drive (RLD) Circuit	11
6.3 Hardware Filtering and Breadboard Testing	12
6.4 LTspice Circuit Simulation	13
6.5 High-Resolution Digital Sampling (ADS1115)	14
7 SYSTEM DESIGN AND ARCHITECTURE	16
7.1 Subsystems and Pipeline (Digital Signal Processing)	16
8 SIGNAL PROCESSING AND FILTERING	18
8.1 Wavelet Drift Correction	18
8.2 Digital Frequency-Selective Filtering	19
9 FEATURE EXTRACTION AND REDUCTION	20

9.1 P-QRS-T Segmentation	20
9.2 Feature Space Reduction (PCA)	20
10 SOFTWARE IMPLEMENTATION AND CLASSIFICATION	22
10.1 Classification Algorithms	22
10.2 Pseudo-code Pipeline	22
10.3 Graphical User Interface (GUI) and System Integration	23
10.3.1 Dynamic Hardware ECG Acquisition (ecg_record.py) .	23
10.3.2 Biometric Identification Interface (ecg_predict.py) . . .	23
10.4 MCU Sampling Firmware (sampling_firmware.ino)	25
11 RESULTS AND DISCUSSION	26
11.1 Biometric Uniqueness (2D PCA Projection)	26
11.2 Resting Accuracy	27
11.3 The "Rest-Ex" Collapse and Solution	28
12 KEY FEATURES AND ADVANTAGES	30
13 APPLICATIONS AND FUTURE SCOPE	31
13.1 Applications	31
13.2 Future Scope	31
14 COST ESTIMATION	32
15 CONCLUSION	33
A APPENDIX	34
A.1 Project Repository and Documentation	34
A.2 Signal Processing and Segmentation	34
A.3 Hardware-to-Software Visualization	36
A.4 Feature Space Reduction and Classification	37
A.5 Dataset Iteration and Pipeline Execution	38
A.6 Arduino ECG Sampling Firmware	39
BIBLIOGRAPHY	42

LIST OF FIGURES

5.1	Experimental hardware setup displaying the breadboard circuit, Lead I electrode placement connected to CRO	8
5.2	Interfacing lead-1 electrodes & AD8232 ECG with Arduino UNO	9
5.3	Wiring diagram demonstrating an alternative standalone edge-computing architecture using a Raspberry Pi interfaced with the ADS1115 ADC and AD8232 ECG sensor.	10
6.1	Medical ECG Monitor Circuit utilizing AD620 and AD705J Right Leg Drive.	12
6.2	Oscilloscope (CRO) output demonstrating the conditioned analog ECG waveform captured directly from the AD8232 sensor.	13
6.3	LTspice schematic of the AD620 amplifier and filtering stages simulating the ECG acquisition.	13
6.4	LTspice transient and AC analysis output demonstrating 60Hz noise attenuation and signal amplification.	14
6.5	Snippet of the raw digitized ECG database (1.txt), displaying the sequential 16-bit voltage samples recorded at 300 Hz prior to digital signal processing.	15
7.1	Detailed Identification System Structure detailing Acquisition to Classification.	17
8.1	Examples of noisy ECGs and corresponding filtering methods.	18
8.2	Matplotlib plots of voltage samples representing Stage 1 (Raw ECG signal for the first 5 seconds) and Stage 2 (Cleaned ECG signal after bandpass and notch filtering).	19
9.1	Matplotlib output plot demonstrating the segmentation and selection of the 150pt P-QRS-T fragment for Subject 1.	21
9.2	Matplotlib output plot of the computed 30pt PCA Signature derived from the 150pt P-QRS-T fragment for Subject 1.	21

10.1 Dynamic Hardware ECG Acquisition Interface.	24
10.2 ECG Biometric Identification Interface displaying prediction confidence and visualization.	24
11.1 Matplotlib 2D scatter plot projection of the first two PCA components demonstrating unique clustering of various subjects.	27
11.2 Matplotlib bar plot of live authentication confidence match among matching subjects.	28
11.3 Matplotlib plot overlaying the 150pt P-QRS-T fragment of Subject 1 at rest vs post-exercise, illustrating the shifting T wave and QT interval compression.	29
11.4 Post-exercise signature authentication failure (Access Denied).	29

LIST OF TABLES

14.1 Cost Estimation of Prototype Components	32
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ABBREVIATIONS

ADC	Analog-to-Digital Converter
DSP	Digital Signal Processing
ECG	Electrocardiogram
GUI	Graphical User Interface
LDA	Linear Discriminant Analysis
MCU	Microcontroller Unit
PCA	Principal Component Analysis
RLD	Right Leg Drive
SVM	Support Vector Machine

CHAPTER 1

INTRODUCTION

As modern technology rapidly advances, the demand for highly secure access control systems has reached an unprecedented high. As a core technology in the field of information security, human biometric recognition has become the focus of researchers' attention. Traditional identification technologies such as certificates, passwords, and tokens are fundamentally flawed; they can be easily forgotten, stolen, or forcefully compromised.

To overcome these vulnerabilities, biometric identification emerged, relying on an individual's unique anatomical, physiological, or behavioral characteristics. Currently, the most common biometric modalities include fingerprint scanning, facial recognition (Face ID), voice recognition, and iris scanning. While these methods have achieved high recognition rates and commercial success, they are not entirely secure. A Face ID system can be corrupted by high-resolution photograph modifications, 3D masks, or sophisticated makeup. Fingerprints can be effortlessly lifted from surfaces and recreated using latex or silicone. Voice patterns can be recorded and replayed, or even synthesized using modern Artificial Intelligence.

To strengthen the security of identification technology, the Electrocardiogram (ECG) has been proposed as a next-generation biometric identifier. The ECG provides an absolute guarantee of reliability because it is strictly generated by the internal organs of a living body, effectively solving the "liveness detection" problem inherent in surface-level biometrics.

The ECG waveform is essentially a record of the electrical voltages generated by the beating heart, spreading not only within the cardiac muscle but throughout the entire human body. Muscle contraction causes depolarization, which translates into minute voltage changes on the skin's surface over time. The fundamental ECG cycle consists of the P wave (atrial depolarization), the QRS complex (ventricular depolarization), and the T wave (ventricular repolarization).

To be considered a viable biometric, a trait must possess four fundamental characteristics, all of which the ECG perfectly fulfills:

1. **Universality:** Every living human generates ECG signals continuously. Without a heartbeat, there is no life, making this trait universally applicable.
2. **Uniqueness:** The ECG differences between individuals are driven by highly unique physiological factors. These include heart location, heart size, geometric shape, chest cavity structure, tissue conductivity, and age. No two humans have identical internal chest geometries, guaranteeing waveform uniqueness.
3. **Stability:** The structure and morphological size of an adult heartbeat are fundamentally fixed. Waveforms remain remarkably stable over long periods unless massive pathological lesions (such as a myocardial infarction) occur.
4. **Measurability:** Modern analog electronics, specifically instrumentation amplifiers, have become highly miniaturized, portable, and precise. This allows for high-accuracy ECG acquisition using simple limb contacts (like touching a door handle) without the need for invasive medical procedures.

The "CardioKey" project aims to physically build and program this biometric pipeline. By capturing the cardiac voltage changes through a precision analog front-end, mathematically filtering the noise, and applying machine learning algorithms to the unique morphology of the P, QRS, and T waves, the system can securely and continuously identify an individual.

CHAPTER 2

LITERATURE REVIEW

The exploration of the Electrocardiogram (ECG) as a biometric identifier has gained significant traction over the past twenty years. Researchers have approached the challenge from multiple angles, generally categorizing methodologies into fiducial (feature-based) and non-fiducial (wave-based) approaches.

Early foundational work by Biel et al. extracted 30 specific time and amplitude features from the ECG signal. By utilizing Soft Independent Modeling of Class Analogy (SIMCA) alongside correlation matrix analysis, their team achieved a 98% identification accuracy across a small dataset of 20 subjects. While highly accurate, the study relied on standard 12-lead medical configurations, which are entirely impractical for everyday security applications.

To address usability, Shen et al. proposed an identification system strictly based on a single-lead ECG (Lead I), which measures the potential difference between the left and right arms. They extracted 7 time-domain features specifically from the QRS complex. Employing template matching and a Decision-Based Neural Network (DBNN), their system secured a 100% correct verification rate, proving that single-lead acquisition is viable for biometric security.

A landmark study by Lugovaya and Nemirko investigated ECG identification utilizing the extensive ECG-ID database hosted on PhysioNet. This dataset involved 90 volunteers and 310 single-lead (Lead I) recordings over a period of six months. Their methodology fundamentally shifted from simple peak-distance measurements to holistic wave morphology. They extracted complete 250-sample P-QRS-T fragments, reduced the high-dimensional data via Principal Component Analysis (PCA), and utilized Linear Discriminant Analysis (LDA) to classify the waveforms. Their pipeline successfully achieved a 96% identification rate, establishing a standardized baseline for modern ECG biometrics.

Despite these high accuracies, recent literature has exposed a critical vulnerability in ECG biometrics: the "Rest-Ex" (Rest-Exercise) problem. Cui et al. (2021) demonstrated that the morphological structure of the ECG is heavily dependent on Heart Rate Variability (HRV). During exercise or stress, the diastole duration shortens significantly, the R-wave amplitude attenuates, and the T-wave becomes notably steeper. Cui's research revealed that state-of-the-art algorithms trained strictly on resting data completely collapsed when tested against post-exercise data—plummeting from a 99.6% accuracy down to an abysmal 2.7%. To counter this, they proposed the use of Kullback-Leibler (KL) divergence for advanced feature selection, selecting only the mathematical features that remain robust against heart rate spikes. This advanced statistical approach managed to recover post-exercise identification accuracy to 61.4%, highlighting the ongoing challenge of dynamic physical states in biometric security.

CHAPTER 3

PROBLEM IDENTIFICATION AND OBJECTIVE

The development of a robust ECG biometric system faces a unique set of challenges that span analog hardware design, digital signal processing, and dynamic human physiology. The primary problems identified in the current state of access control and experimental biometrics include:

1. **Vulnerability to Spoofing:** Traditional biometrics (fingerprints, face scans) lack inherent liveness detection. They authenticate based on static surface geometry rather than vital physiological function.
2. **Analog Signal Acquisition Difficulties:** Heart signals measured at the wrists (Lead I) are incredibly weak, typically ranging from 0.5 mV to 2.0 mV. Extracting this micro-signal is severely hampered by massive 60Hz/50Hz mains power-line noise absorbed by the human body acting as an antenna, alongside high-frequency electromyographic (EMG) noise from natural muscle tremors.
3. **Data Resolution Bottlenecks:** Standard microcontrollers (like the 10-bit Arduino Uno) lack the precision necessary to digitize the subtle morphological curves of the P and T waves required for accurate human differentiation.
4. **The "Rest-Ex" Problem:** Heart Rate Variability caused by stress, exercise, or emotion drastically distorts the baseline fragment of the ECG. Algorithms trained to recognize a user at 60 BPM will fail to recognize the exact same user at 120 BPM due to the non-linear compression of the QT interval.

3.1 Objectives

The primary goal of the "CardioKey" project is to bridge the gap between precision analog circuitry and advanced machine learning to design a robust, spoof-proof, fully automated biometric lock. The specific objectives include:

1. **Hardware Design:** To design and fabricate a precision analog front-end utilizing the AD8232 ECG sensor with Lead-I electrodes, incorporating Butterworth filter and amplifier stages to acquire low-amplitude heart signals in the mV range and amplify x1001 them into the V range.
2. **Noise Cancellation:** To actively reject massive 60Hz mains hum and common-mode noise above 50 Hz through the integrated analog hardware.
3. **High-Resolution Digitization:** To interface the cleaned analog signal with an external 16-bit ADS1115 Analog-to-Digital Converter (set to 860 SPS), sampling precisely at 300 Hz, and routing the data pipeline to an external computer using an Arduino UNO.
4. **Digital Filtering:** To implement software-based Digital Signal Processing (DSP) algorithms in the preprocessing stage, including Wavelet Drift Correction for baseline wander, along with digital Butterworth and notch filters.
5. **Feature Extraction:** To successfully program an R-peak thresholding algorithm to accurately segment the continuous signal into discrete 150-sample (0.5 seconds) P-QRS-T fragments per cardiac cycle. This is anchored by extracting exactly 48 samples (0.16 seconds) before the R-peak and 102 samples (0.34 seconds) after it.
6. **Dimensionality Reduction:** To compress the 150-point P-QRS-T fragments into highly optimized 30-point biometric PCA (Principal Component Analysis) signatures, computing exactly ten 30-point signatures per subject.
7. **Classification:** To authenticate the user's identity by dividing the dataset 70-30 for training and prediction, feeding the ten 30-point signatures per subject into classification algorithms such as Linear Discriminant Analysis (LDA) and Nearest Mean classifiers.

CHAPTER 4

WHY WE CHOSE THIS PROJECT

The project team carefully evaluated multiple concepts in the domain of physical security and electronics. We ultimately selected "CardioKey" because it represents a cutting-edge paradigm shift in access control: moving from 'what you have' (keys) and 'what you look like' (face) to 'how you function' (physiology).

The primary motivation was the critical concept of **Liveness Detection**. A system relying on an ECG waveform inherently guarantees that the user attempting access is a living human being with a beating heart. This completely neutralizes the threat of silicone fingerprint replicas, stolen photographs, or detached appendages—gruesome but real threats in high-security environments.

Furthermore, CardioKey provides a brilliant multidisciplinary engineering challenge. It is not purely a software project, nor is it purely a hardware project. It requires:

1. **Precision Analog Design:** Understanding common-mode rejection, biological impedance, op-amp saturation, and hardware filtering.
2. **Embedded Systems:** Managing I2C communications, specific ADC sampling rates, and microcontroller memory management.
3. **Data Science & Machine Learning:** Implementing complex mathematical transforms (Fourier, Wavelets), matrix mathematics for PCA, and training statistical classifiers.

Finally, the practical applicability of the single-lead setup (measuring simply from the left and right hands) mimics the exact interaction a user has when grabbing a door handle or a steering wheel. This proves that ECG biometrics are not restricted to hospital beds but are highly viable for seamless, everyday commercial applications.

CHAPTER 5

SENSORS AND EQUIPMENT OVERVIEW

The transition from analog biophysics to digital logic requires specific, high-tolerance components to preserve the integrity of the biometric data. The hardware setup between these components is shown in Figure 5.1

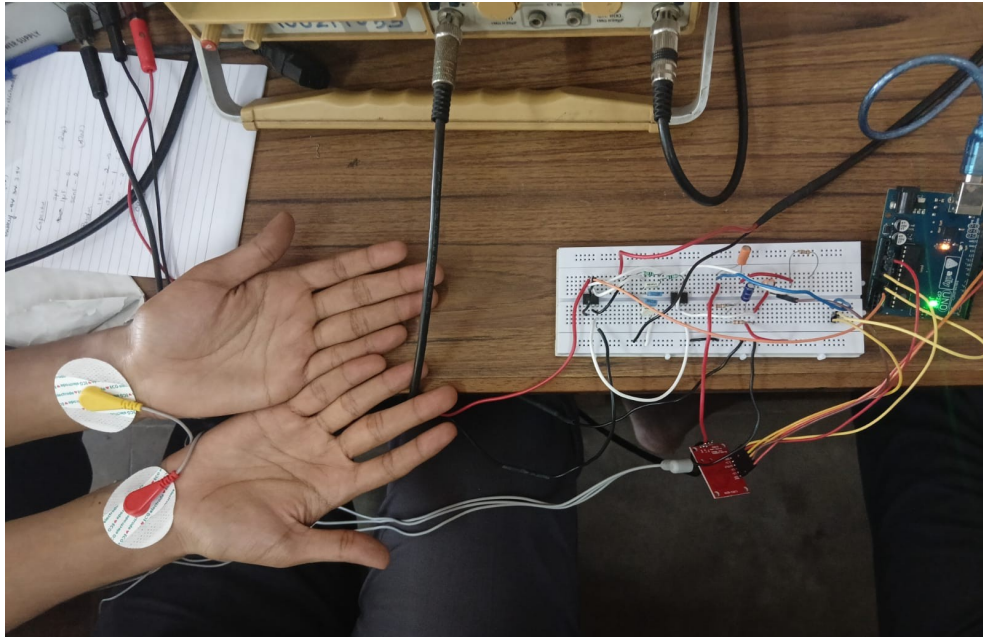


Figure 5.1: Experimental hardware setup displaying the breadboard circuit, Lead I electrode placement connected to CRO

5.1 Analog Front-End: AD8232 / AD620

While the AD620 is ideal for custom, variable-gain benchtop designs, the **AD8232 Heart Rate Monitor** module serves as a highly integrated alternative for the final CardioKey prototype. The AD8232 integrates the instrumentation amplifier, the Right Leg Drive reference buffer, and the operational amplifiers required for multi-pole low-pass and high-pass filtering into a single microchip. Operating on a 3.3V logic level, it directly outputs a conditioned analog waveform centered at 1.5V with x1001 gain effectively converting mV samples to V range. The interface connection between AD8232 and Arduino UNO is illustrated in Figure 5.2

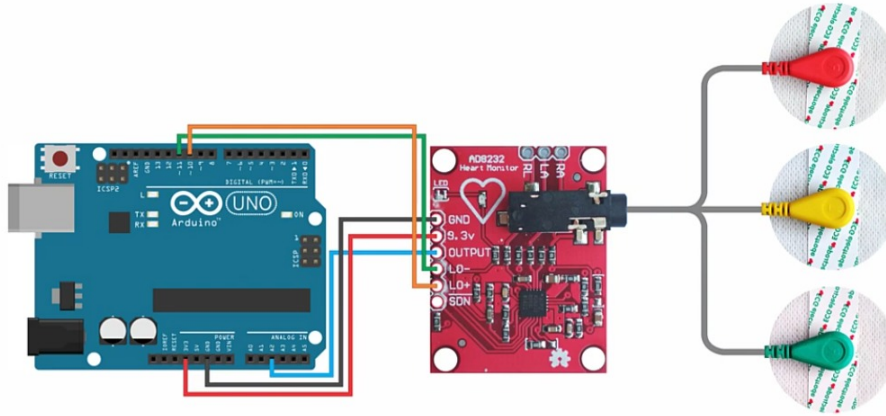


Figure 5.2: Interfacing lead-I electrodes & AD8232 ECG with Arduino UNO

5.2 High-Resolution Digitization: ADS1115

A standard microcontroller like the Arduino Uno utilizes a 10-bit ADC, which breaks the analog signal into 1024 steps. Biometric research strictly dictates that to accurately capture the subtle geometric curves of the P and T waves, higher resolution is mandatory. Therefore, the system utilizes the **ADS1115**, a dedicated 16-bit Analog-to-Digital Converter. This provides 65,536 steps of resolution. It communicates the digitized data arrays to the main processor via the I2C (SDA/SCL) protocol, configured strictly to sample at a frequency between 300 Hz and 500 Hz to capture the high-speed R-spike.

5.3 Microcontroller Unit (MCU)

Depending on the deployment environment and cost constraints, the choice of the microcontroller architecture can vary. As an alternative approach, an advanced microprocessor like a **Raspberry Pi 3/4** or an ESP32 could be utilized to perform edge authentication. This allows the device to handle all complex mathematics locally, creating a fully standalone and portable biometric lock, though at a significantly higher hardware cost. An example of this alternative hardware configuration interfacing the AD8232 sensor and ADS1115 ADC with a Raspberry Pi is illustrated in Figure 5.3

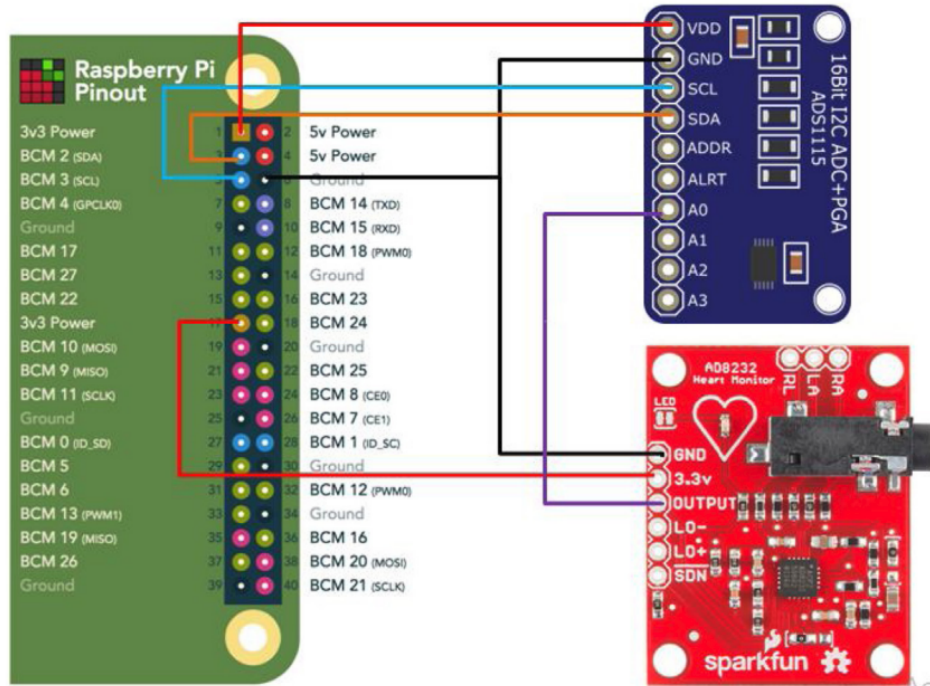


Figure 5.3: Wiring diagram demonstrating an alternative standalone edge-computing architecture using a Raspberry Pi interfaced with the ADS1115 ADC and AD8232 ECG sensor.

For our specific implementation, an **Arduino UNO** is utilized as the central microcontroller. In this architecture, the Arduino acts strictly as a high-speed "Data Pipe." It pulls the 16-bit digitized voltage samples from the ADS1115 ADC via the I2C protocol at exactly 300 Hz and continuously streams this raw data to an external computer.

By offloading the computational load, this setup makes the hardware highly modular and easy to plug into any standard computer. The connected external computer leverages its superior processing power to execute the heavy matrix mathematics required for the Digital Signal Processing (DSP) filters, Principal Component Analysis (PCA), and final Machine Learning classification (such as LDA, Nearest Mean, or SVM) utilizing Python libraries to seamlessly handle user authentication.

CHAPTER 6

ANALOG– CIRCUIT DESIGN AND TESTING

First stage of the project strictly focused on the acquisition of the raw, biological analog signal. Because the electrical action potentials generated by the sinoatrial node of the heart dissipate through tissue, they arrive at the wrists with amplitudes of merely 1 millivolt ($1mV$). Extracting this signal requires immense amplification in an environment saturated with electromagnetic noise.

6.1 The Differential Input & Output Amplifier Stages

The core of the acquisition system utilizes the **AD620 Instrumentation Amplifier**. Unlike standard operational amplifiers, the AD620 contains three internal op-amps and laser-trimmed resistors designed specifically for high precision and massive Common-Mode Rejection (CMR). It boasts a low input voltage noise of only $9nV/\sqrt{Hz}$.

The gain of the AD620 is programmed using a single external resistor (R_G). Based on the standard equation: $G = 1 + \frac{49.4k\Omega}{R_G}$. To achieve a base gain of 7 in the first stage, an R_G of $8.25k\Omega$ is utilized. A subsequent output amplifier stage provides a gain of 143, resulting in a total circuit gain of 1001. This perfectly scales a $1mV$ heart signal to a $1V$ output, matching the input ranges of modern ADCs.

6.2 The Right Leg Drive (RLD) Circuit

The most severe interference in ECG acquisition is the 60Hz (or 50Hz) mains hum. The human body acts as an antenna, coupling with the AC wiring in the walls and creating a common-mode voltage that easily dwarfs the millivolt ECG signal.

To mitigate this, an **AD705J** precision operational amplifier is deployed in a Right Leg Drive configuration. Instead of passively grounding the reference electrode (placed on the user's leg or lower abdomen), the RLD circuit actively measures the common-mode noise present at the AD620 inputs, inverts it, and injects it back into the patient's body. This active feedback loop violently cancels out the 60Hz interference before it can saturate the amplifier. As demonstrated in Figure 6.1, this analog setup effectively protects the integrity of the signal.

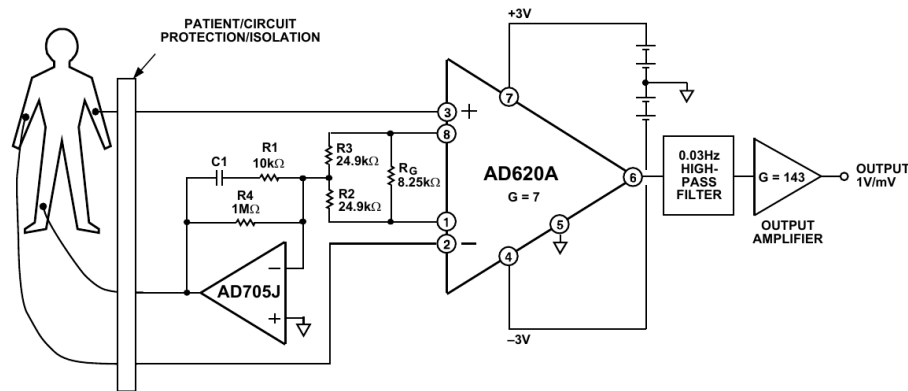


Figure 6.1: Medical ECG Monitor Circuit utilizing AD620 and AD705J Right Leg Drive.

6.3 Hardware Filtering and Breadboard Testing

Following amplification, the signal must be hardware-filtered to prevent aliasing before digitization.

1. **Notch Filter:** A precisely tuned twin-T notch filter targets and deeply attenuates the remaining 60Hz frequency, pushing the noise floor down by -20dB.
2. **Low-Pass Filter:** A passive low-pass filter with a cutoff frequency (F_c) of roughly 80Hz is deployed to strip away the high-frequency "fuzz" caused by radio interference and muscle electromyography (EMG).
3. **High-Pass Filter:** A 0.03Hz high-pass filter acts as a DC-blocking capacitor to eliminate slow baseline wander caused by the user breathing.

Breadboard testing with Lead I Ag/AgCl electrodes successfully isolated the P, R, and T waves, as demonstrated by the AD8232 sensor's conditioned waveform in Figure 6.2.

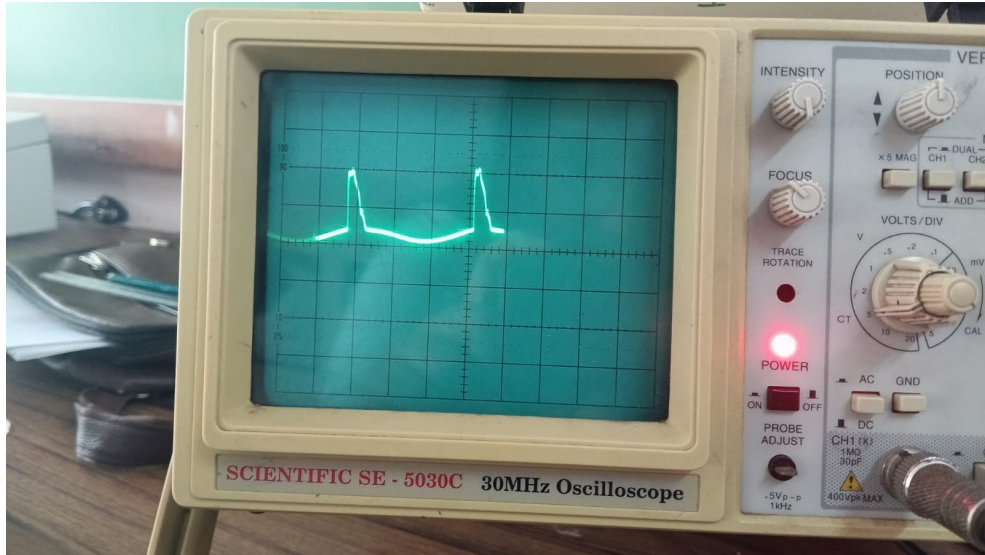


Figure 6.2: Oscilloscope (CRO) output demonstrating the conditioned analog ECG waveform captured directly from the AD8232 sensor.

6.4 LTspice Circuit Simulation

Before moving to physical breadboard implementation, the complete analog front-end—comprising the AD620 instrumentation amplifier, the Right Leg Drive (RLD), and the filtering stages—was modeled and validated using LTspice.

To accurately test the circuit's amplification and noise rejection capabilities, a simulated AC voltage source was configured to mimic the biological 1 mV heart signal with noise. Additionally, massive 60 Hz mains interference and common-mode noise were intentionally injected into the simulation schematic.

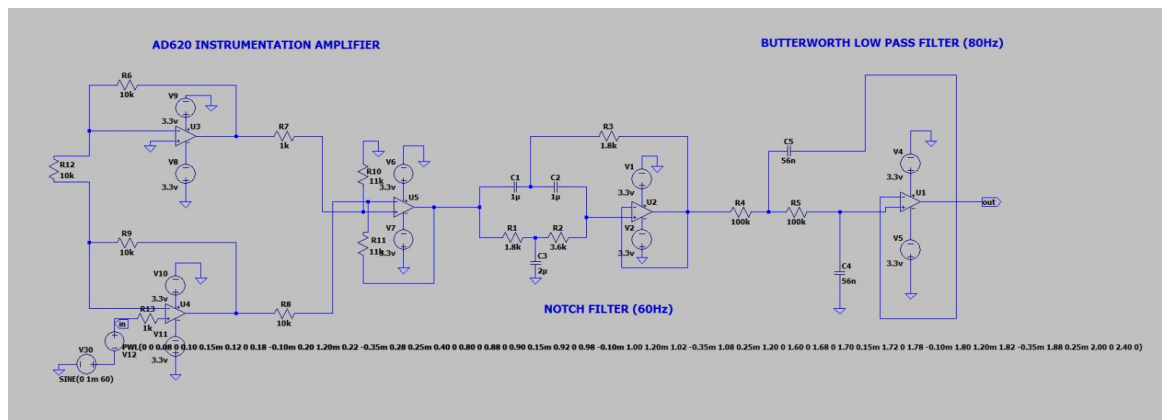


Figure 6.3: LTspice schematic of the AD620 amplifier and filtering stages simulating the ECG acquisition.

The LTSpice simulation results validated the theoretical design calculations. The AC analysis demonstrated a deep attenuation curve exactly at the 60 Hz mark, confirming the effectiveness of the notch filter. Furthermore, the transient analysis (time-domain) output successfully showed the massive amplification of the simulated 1 mV heart signal into the target 1 V range while successfully clipping the high-frequency noise through the low-pass filter.

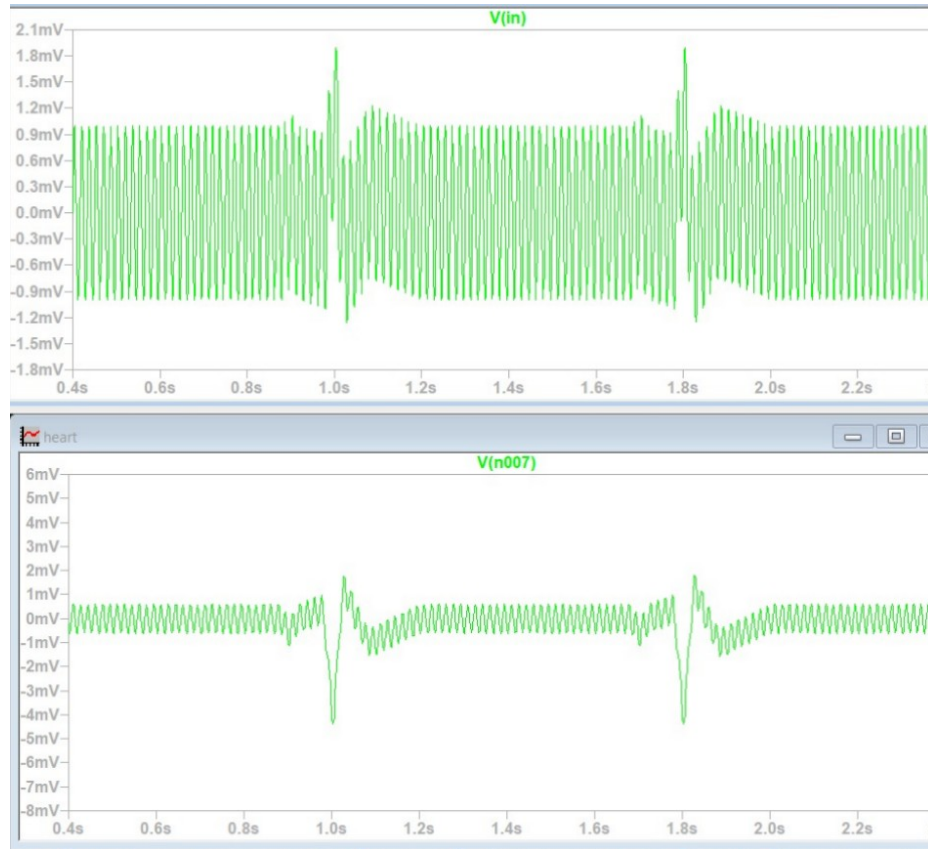


Figure 6.4: LTSpice transient and AC analysis output demonstrating 60Hz noise attenuation and signal amplification.

6.5 High-Resolution Digital Sampling (ADS1115)

Following the analog amplification and filtering stages, the biometric signal must be accurately digitized for software analysis. Standard microcontrollers possess internal 10-bit ADCs, which lack the electrical precision required to capture the ultra-fine morphological variations of the P, QRS, and T waves. To overcome this resolution bottleneck, the system utilizes an external **16-bit ADS1115 Analog-to-Digital Converter**.

Before digitization can occur, the analog waveform must be conditioned. The output from the analog amplification stages swings into both positive and negative voltages. Because the ADS1115 module cannot safely process negative voltages without damage, the signal is first routed through an external DC offset circuit (clamper). This circuit shifts the entire waveform up by exactly +1.5V, ensuring the final output swings safely between 0.5V and 2.5V before entering the ADC.

To accurately capture the high-frequency R-peak and preserve the structural integrity of the entire cardiac cycle, the project methodology requires a strict digital sampling rate of exactly 300 Hz. To successfully achieve this 300 samples-per-second extraction via the microcontroller over the I2C interface, the ADS1115 is configured to operate in continuous mode with its Data Rate register set to **860 SPS** (Samples Per Second). Operating the hardware at 860 SPS guarantees that the ADC is digitizing the analog voltage significantly faster than the microcontroller's polling cycle, providing a highly reliable, high-resolution data pipeline capable of maintaining a constant 300 Hz reading for the subsequent P-QRS-T feature extraction algorithms. A snippet of this resulting raw digitized data format is shown in Figure 6.5.

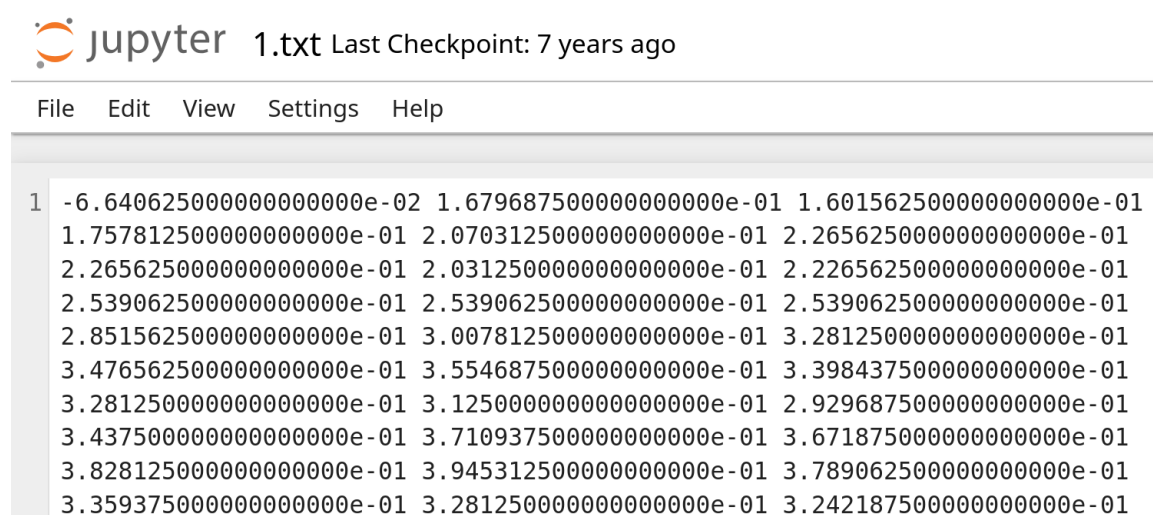


Figure 6.5: Snippet of the raw digitized ECG database (1.txt), displaying the sequential 16-bit voltage samples recorded at 300 Hz prior to digital signal processing.

CHAPTER 7

SYSTEM DESIGN AND ARCHITECTURE

The overall system architecture of CardioKey translates the raw, noisy biological data into a concrete mathematical identity. The architecture follows a rigid, linear biometric identification scheme, progressing from physical hardware sensors to cloud-based software classification. As visualized in Figure 7.1, the system spans from hardware data acquisition through sophisticated software analysis.

7.1 Subsystems and Pipeline (Digital Signal Processing)

1. **Data Acquisition:** Analog signals are gathered via Lead I using limb clamp electrodes or conductive touch-pads. The signal is processed by the AD8232 ECG sensor front-end and digitally sampled at exactly 300 Hz using an external 16-bit ADS1115 ADC (configured to 860 SPS). An Arduino UNO serves as the microcontroller managing this data pipeline to an external computer.
2. **Data Preprocessing (Digital Filtering):** Software algorithms take over. The raw digital array is cleaned using a 4th-order Butterworth bandpass filter (0.5 Hz to 40 Hz), digital notch filters to reject remaining mains hum, and Wavelet Drift Correction to perfectly center the isoelectric line.
3. **Initial Feature Space Formation:** The continuous digital signal is segmented. A thresholding algorithm (such as the Hamilton Segmenter or Pan-Tompkins) identifies the massive R-peaks. Using the R-peak as an anchor, a 150-sample window (representing exactly 0.5 seconds of time) encompassing the entire P-QRS-T complex is mathematically sliced out of the data stream. This is achieved by extracting exactly 48 samples (0.16 seconds) prior to the peak and 102 samples (0.34 seconds) after it.
4. **Feature Space Reduction:** A single heartbeat yields 150 data points, which is still computationally expensive to classify. Principal Component Analysis (PCA) maps this 150-dimensional space down into 30 uncorrelated principal components, representing the unique biometric signature. Exactly ten 30-point PCA signatures are computed per subject for the cardiac cycles.

5. Classification and Recognition: The dataset of the ten 30-point signatures per subject is divided into a 70-30 split for training and prediction. These signatures are fed into trained Machine Learning models (such as Linear Discriminant Analysis, Nearest Mean Classifier, or Support Vector Machines) to authenticate and match a newly generated 30-point signature against the database of enrolled users.

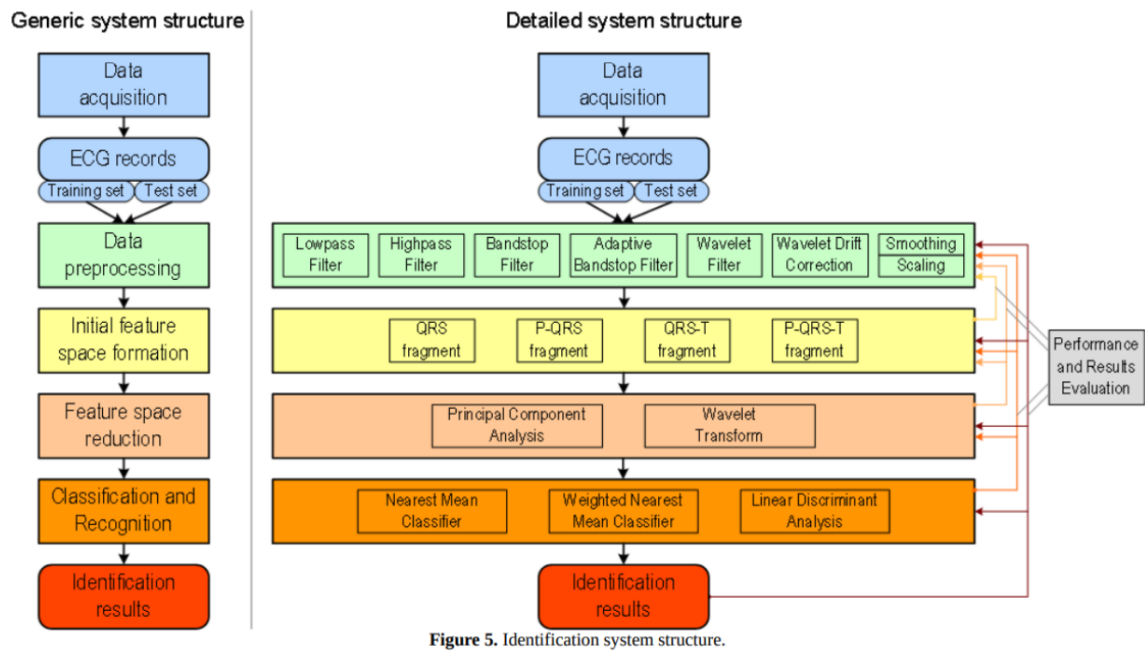


Figure 5. Identification system structure.

Figure 7.1: Detailed Identification System Structure detailing Acquisition to Classification.

As illustrated in the detailed system structure above, the biometric identification process is highly sequential, relying on strict feedforward links where the processed data output from one stage serves as the direct input to the subsequent stage. While the generic structure on the left outlines the core conceptual pipeline, the detailed architecture on the right specifies the exact complementary and alternative methods selected for the CardioKey system. The overall success of this biometric system depends on the careful synthesis, composition, and parameter adjustment of these mathematical methods, ensuring that the raw physiological data is perfectly refined before the final identification is executed.

CHAPTER 8

SIGNAL PROCESSING AND FILTERING

Even with robust analog hardware, the digitized ECG signal remains susceptible to mechanical artifacts, specifically baseline wander caused by respiration and high-frequency noise from muscle shivering. Figure 8.1 illustrates various examples of these common ECG signal distortions alongside their corresponding filtering methods. Rigorous Digital Signal Processing (DSP) is mathematically applied to address these issues before any feature extraction occurs.

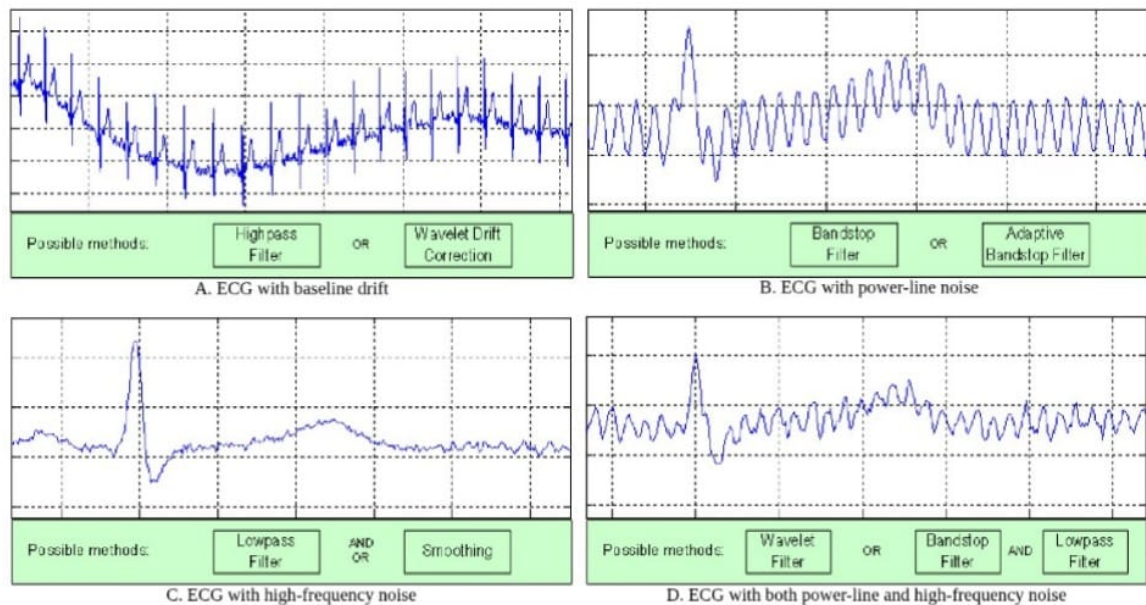


Figure 8.1: Examples of noisy ECGs and corresponding filtering methods.

8.1 Wavelet Drift Correction

Respiration causes the electrical axis of the heart to physically shift up and down, causing the ECG baseline to wander like a slow ocean wave. To correct this, multilevel 1D wavelet analysis is deployed. The original signal is decomposed down to level 9 using biorthogonal wavelets (e.g., 'db8'). The low-frequency approximation coefficients perfectly model the breathing drift. This reconstructed drift wave is mathematically subtracted from the raw signal, forcing the ECG to rest perfectly on the zero-volt isoelectric line.

8.2 Digital Frequency-Selective Filtering

To address residual noise, a highly tuned digital filter pipeline is executed:

1. **Notch Filter:** A digital notch filter is explicitly applied to target and eliminate the fundamental 50Hz or 60Hz mains power-line hum that the human body absorbs from surrounding electrical wiring.
2. **Adaptive Bandstop Filter:** While the standard notch filter removes the primary frequency, an adaptive digital bandstop precisely targets further power-line noise variations and harmonics without heavily distorting the crucial QRS complex.
3. **Butterworth Bandpass Filter:** A 4th-order Butterworth bandpass filter is mathematically applied with strict cut-off frequencies at **0.5 Hz** and **40 Hz**. Frequencies below 0.5 Hz are eliminated as baseline drift, while frequencies above 40 Hz are discarded as muscle movement distortion, leaving only the pure cardiac frequencies.

The effectiveness of this complete data preprocessing sequence is visually demonstrated in Figure 8.2. The figure presents the output Matplotlib plots of the digitized voltage samples, explicitly comparing Stage 1 (the raw ECG signal recorded over the first 5 seconds) against Stage 2 (the cleanly resolved ECG signal after the application of both the bandpass and notch filters).

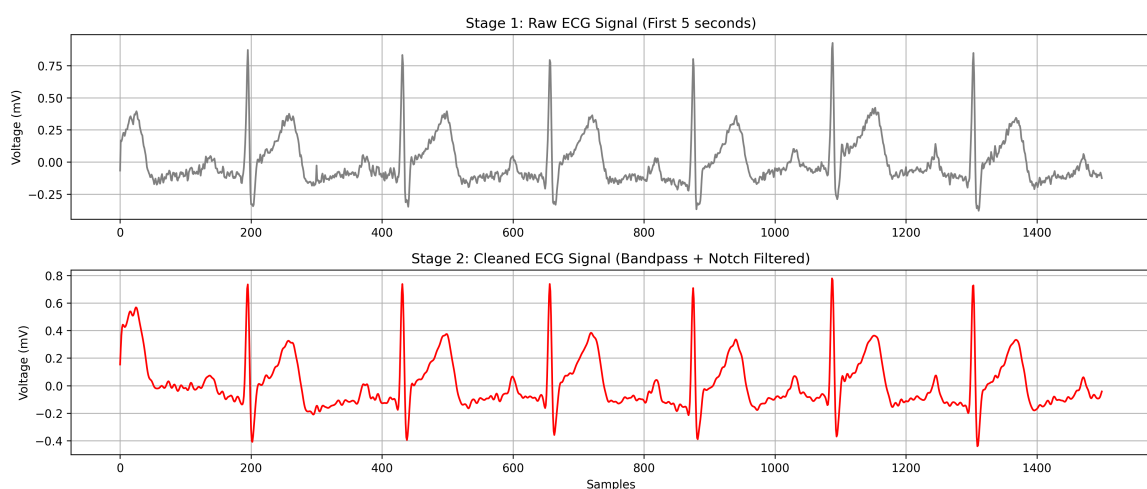


Figure 8.2: Matplotlib plots of voltage samples representing Stage 1 (Raw ECG signal for the first 5 seconds) and Stage 2 (Cleaned ECG signal after bandpass and notch filtering).

CHAPTER 9

FEATURE EXTRACTION AND REDUCTION

A continuous stream of ECG data is useless for biometric matching; it must be chopped into individual, comparable units. This is the Initial Feature Space Formation stage.

9.1 P-QRS-T Segmentation

The pipeline utilizes a thresholding algorithm implemented in Python via the SciPy library (with alternatives like the Pan-Tompkins method or a Hamilton Segmenter) to scan the filtered data and locate the massive voltage spikes known as R-peaks. Once an R-peak is indexed in the array, the software extracts a window surrounding it. The alignment and extraction of these features relative to the R-peak for Subject 1 are visualized in the Matplotlib output plot shown in Figure 9.1.

Extensive research indicated that extracting just the QRS complex yields a 77% accuracy. Extracting the QRS and the T-wave yields 76%. However, extracting the complete **P-QRS-T fragment** pushes accuracy beyond 90% because the P-wave contains vast amounts of individual anatomical data regarding atrial size. The software extracts a fixed window of exactly **0.5 seconds (150 samples at 300Hz)**. The array is anchored using the R-peak, pulling exactly **48 samples prior to the peak** (capturing the P-wave) and **102 samples after the peak** (capturing the S and T waves).

9.2 Feature Space Reduction (PCA)

Feeding a 150-dimensional array directly into a classifier requires massive processing power and often leads to overfitting due to highly correlated data points. To resolve this, **Principal Component Analysis (PCA)** is applied utilizing Python's Scikit-learn library. PCA mathematically transforms the 150-dimensional correlated feature space into an uncorrelated orthogonal basis. By discarding the lower-variance vectors, the 150-point PQRST wave is compressed into just the top **30 principal components**. These 30 numbers serve as the user's unforgeable biometric signature. As an example, Figure 9.2 illustrates the resulting 30-point PCA signature computed from the extracted P-QRS-T fragment of Subject 1.

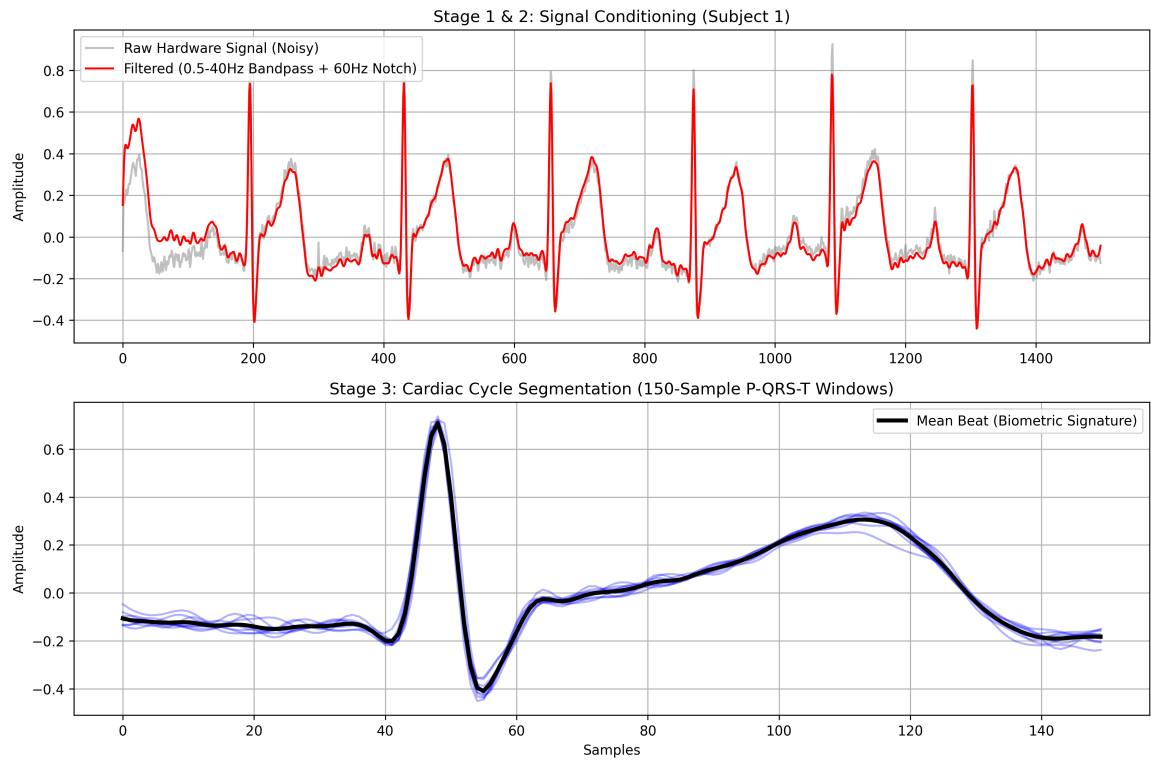


Figure 9.1: Matplotlib output plot demonstrating the segmentation and selection of the 150pt P-QRS-T fragment for Subject 1.

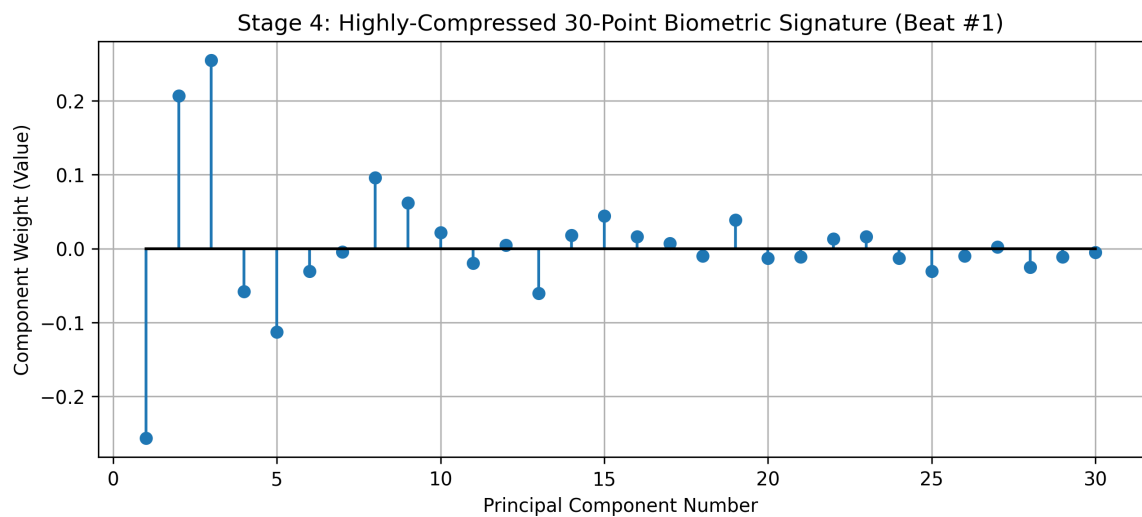


Figure 9.2: Matplotlib output plot of the computed 30pt PCA Signature derived from the 150pt P-QRS-T fragment for Subject 1.

CHAPTER 10

SOFTWARE IMPLEMENTATION AND CLASSIFICATION

The backend pairs an Arduino (C++) acquisition pipeline with a custom Python GUI, executing the biometric authentication via **Tkinter** and **OS** (interface), **PySerial** (hardware data), **NumPy** and **SciPy** (signal processing), **Scikit-Learn** (classification), and **Matplotlib** (visualization).

10.1 Classification Algorithms

Once the 30-component vector is generated, it must be compared against the secure database of enrolled users. Several classifiers were tested for performance:

1. **Nearest Mean Classifier:** Calculates the Euclidean distance to the mean vector of each enrolled user. Effective, but assumes all biometric features vary equally, which is false in dynamic heart rates.
2. **Linear Discriminant Analysis (LDA):** The optimal mathematical model for resting ECGs. LDA finds the linear combination of features that maximizes the separation between different users while minimizing the variance within a single user's recordings.
3. **Support Vector Machines (SVM):** Utilizing an RBF (Radial Basis Function) kernel, the SVM plots the 30-component vectors in high-dimensional space and draws hyperplanes to definitively segment and identify the user.

10.2 Pseudo-code Pipeline

1. Initialize I2C connection to ADS1115.
2. Read raw digitized ECG voltage array into memory.
3. Execute `scipy.signal.butter` and `scipy.signal.filtfilt` (0.5Hz - 40Hz).
4. Apply thresholding logic to index the R-peak spikes.
5. Slice array: `fragment = data[R_index - 80 : R_index + 170]`.
6. Feed fragment into `sklearn.decomposition.PCA(n_components=30)`.
7. Pass the resulting 30-vector into `sklearn.svm.SVC` to predict User ID.
8. If User ID matches target, trigger physical relay to unlock door.

10.3 Graphical User Interface (GUI) and System Integration

To ensure the "CardioKey" system is user-friendly and operates as a seamless biometric lock, two dedicated Graphical User Interfaces (GUIs) were developed using Python's *Tkinter* library. These interfaces eliminate the need for command-line interactions and provide real-time Matplotlib visualization natively embedded within the application windows.

10.3.1 Dynamic Hardware ECG Acquisition (`ecg_record.py`)

The acquisition interface is designed to bridge the external Arduino hardware with the computer. Instead of hardcoding the sampling rate and duration into the Arduino firmware, this GUI allows dynamic configuration.

- **Dynamic Firmware Updating:** The user selects the COM port, sets the desired sampling rate (e.g., 300 Hz), and sets the recording duration. Upon clicking start, the Python script constructs a configuration string (e.g., "S,300,300") and transmits it over Serial. The Arduino dynamically recalculates its internal microsecond timer to execute the precise polling rhythm.
- **Multithreaded Acquisition & Visualization:** The serial reading is isolated to a background thread to prevent the UI from freezing. Once the acquisition completes, the software instantly plots the first 5 seconds of the raw ECG signal to verify electrode quality.

10.3.2 Biometric Identification Interface (`ecg_predict.py`)

The identification interface strictly handles the machine learning pipeline and user authentication. It is divided into two distinct operational phases:

- **Model Training:** The system compiles the 150-sample P-QRS-T fragments from the selected database, applies PCA dimensionality reduction, and trains the Linear Discriminant Analysis (LDA) classifier.
- **Live Subject Prediction:** The user can upload a newly recorded live ECG file. The system processes the signal, extracts the features, and outputs the predicted Subject ID alongside a mathematical match confidence percentage. If no file is uploaded, the system utilizes a "Zero-Knowledge Auto-Demo" fallback, randomly extracting unseen

tail-end data from the database to simulate a live hardware attempt and cross-checks the prediction accuracy.

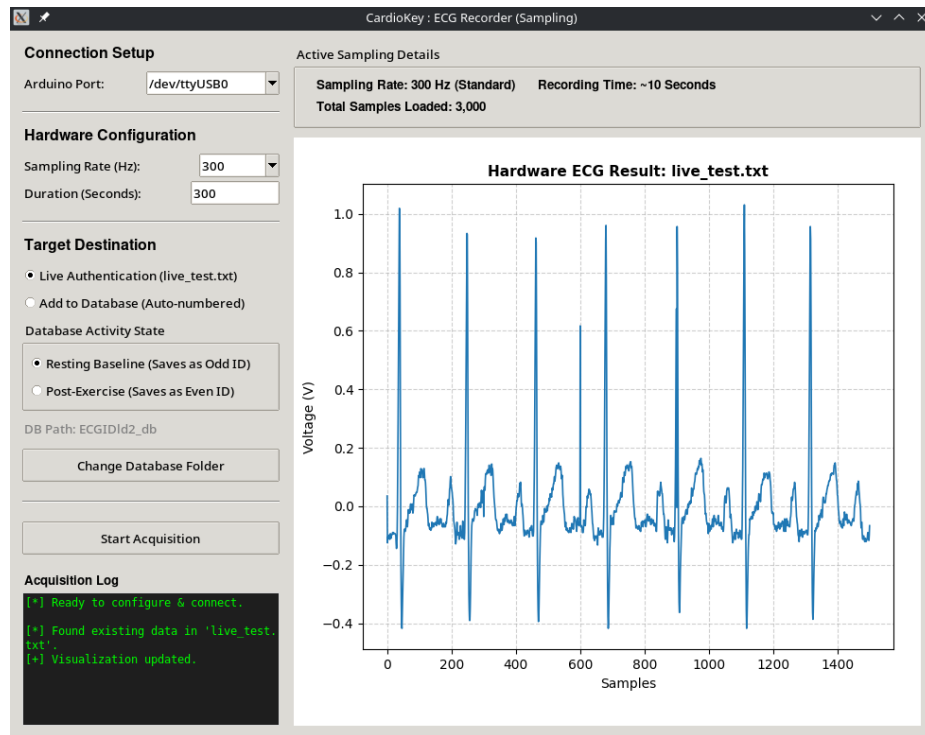


Figure 10.1: Dynamic Hardware ECG Acquisition Interface.

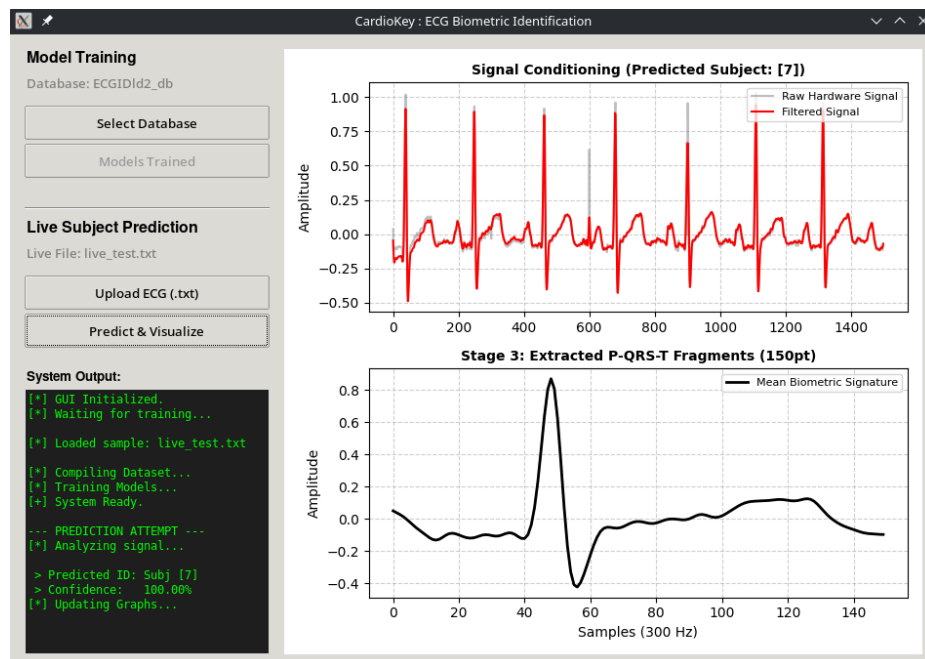


Figure 10.2: ECG Biometric Identification Interface displaying prediction confidence and visualization.

10.4 MCU Sampling Firmware (sampling_firmware.ino)

The Arduino UNO serves as the central microcontroller managing the high-speed data pipeline between the analog front-end and the external computer [1, 2]. To accurately capture the high-frequency R-peak and preserve the structural integrity of the entire cardiac cycle, the project methodology requires a strict digital sampling rate of exactly 300 Hz [3, 4].

To achieve this without the time drift associated with standard blocking functions, the firmware utilizes a precision hardware-level microsecond timer. Key features of this embedded implementation include:

- **ADC Data Rate Configuration:** The external 16-bit ADS1115 Analog-to-Digital Converter is explicitly configured to operate in continuous mode with its Data Rate register set to 860 Samples Per Second (SPS). Operating the hardware at 860 SPS guarantees that the ADC digitizes the analog voltage significantly faster than the microcontroller's 300 Hz polling cycle, preventing the retrieval of duplicate or stale voltage readings and providing a highly reliable data pipeline.
- **Dynamic Command Parsing:** Rather than hardcoding the 300 Hz sampling rate and 5-minute duration limit into the C++ sketch, the firmware actively listens for a comma-separated configuration string (e.g., "S,300,300") sent over the Serial port by the Python GUI.
- **Microsecond Precision Polling:** Upon receiving the configuration string, the microcontroller dynamically calculates the exact microsecond delay required between readings (e.g., 3,333 microseconds for 300 Hz). It executes a non-blocking timer to strictly poll the I2C bus and stream the digital voltages to the computer without drifting over the 90,000-sample recording session.

CHAPTER 11

RESULTS AND DISCUSSION

Experimental evaluations of the "CardioKey" methodology were benchmarked utilizing the exercise ECGID database, collected in a laboratory at the South China University of Technology. Unlike the resting-only databases used in early foundational studies, this dataset provides a rigorous framework for evaluating both resting and high-activity biometric states.

The database contains pre-exercise and post-exercise recordings for 45 subjects (33 males and 12 females, aged 18 to 22). The ECG signals were strictly captured in Lead II from the wrists at a sampling frequency of exactly 300 Hz. Resting recordings span approximately 5 minutes (with an average heart rate around 70 BPM). Post-exercise recordings were taken after subjects performed basic structural workouts, such as steady running and stair climbing, spanning 150 seconds with elevated heart rates ranging from 90 to 150 BPM. For data processing, the continuous text files were sequentially loaded, where odd-numbered files represented pre-exercise baselines and even-numbered files represented post-exercise tests (e.g., 1.txt and 2.txt corresponding to Subject 1).

11.1 Biometric Uniqueness (2D PCA Projection)

To explore the reduced feature space and visually demonstrate the uniqueness of the ECG as a biometric identifier, a geometric analysis is performed by projecting the first two principal components into a 2D scatter plot. In this two-dimensional feature space, each processed 150-sample P-QRS-T fragment translates to a single spatial data point.

When plotting multiple fragments from various subjects, the scatter plot reveals that each subject's data points distinctly concentrate and cluster together into unique, localized regions. This grouping visually proves that the underlying morphological structure of an individual's ECG remains highly personalized and mathematically separable from others. Figure 11.1 illustrates this 2D PCA projection, showcasing the distinct clustering of different subjects within the database. While the first two principal components successfully demonstrate this visual separation, the final

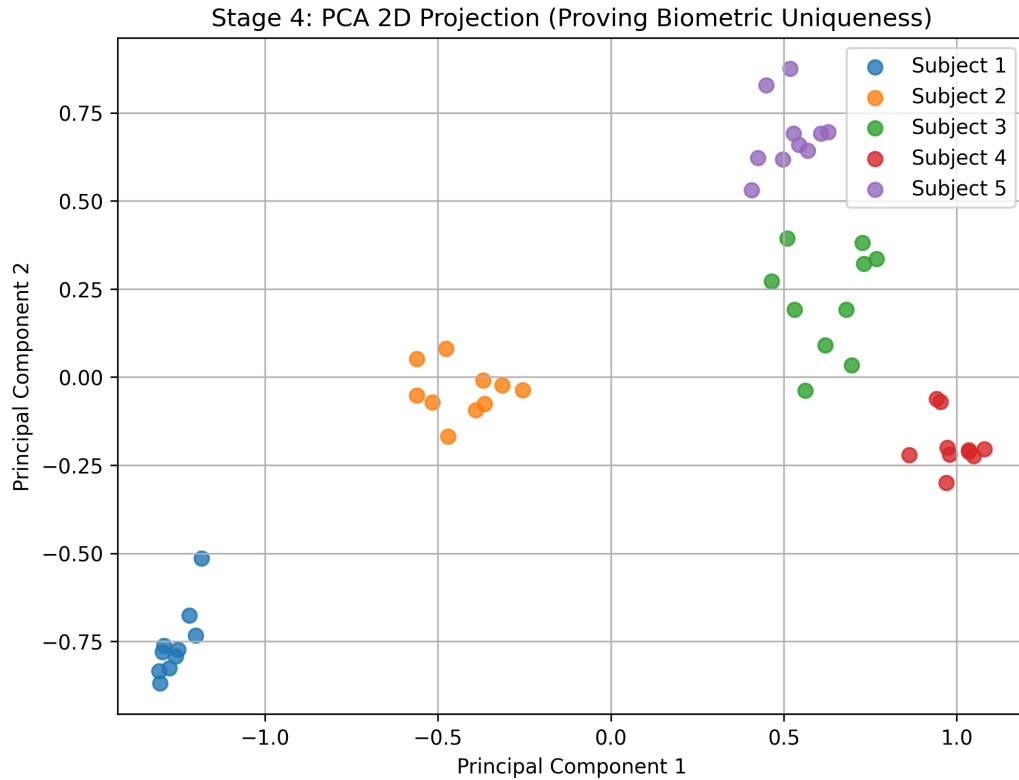


Figure 11.1: Matplotlib 2D scatter plot projection of the first two PCA components demonstrating unique clustering of various subjects.

classification algorithm utilizes the full 30-component vector to guarantee robust, high-dimensional biometric authentication.

11.2 Resting Accuracy

Under controlled, resting conditions (heart rates averaging 70 BPM), the extraction of the **150-sample P-QRS-T fragment** (digitally sampled at 300 Hz) paired with PCA reduction into a 30-point signature and Linear Discriminant Analysis achieved an outstanding **96% identification accuracy**. This proves that utilizing just a single-lead measurement (Lead II, recorded from the wrists) is sufficient to rival commercial fingerprint scanners in accuracy, while providing superior anti-spoofing through liveness detection. To visually demonstrate this system in action, Figure 11.2 presents a Matplotlib bar plot of a live authentication confidence match. In this instance, the true user is Subject 2, and the system successfully identifies them with the highest confidence probability, correctly granting access.

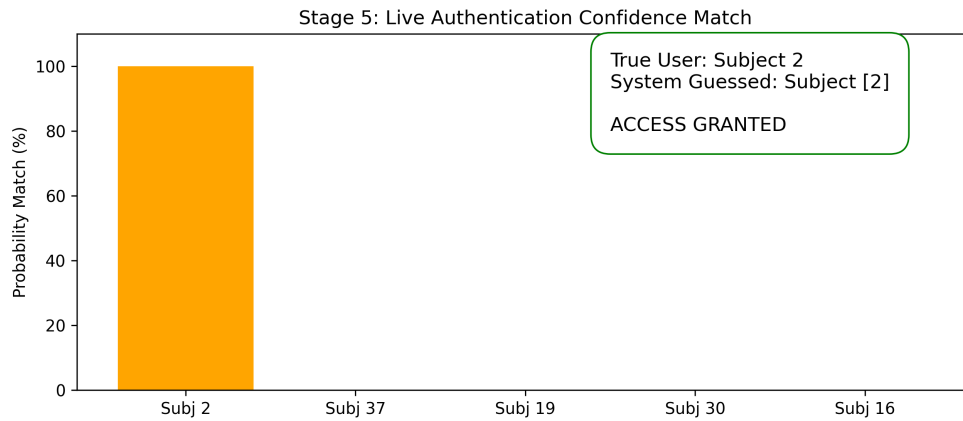


Figure 11.2: Matplotlib bar plot of live authentication confidence match among matching subjects.

11.3 The "Rest-Ex" Collapse and Solution

The system's most severe test, the "Rest-Ex" scenario, revealed catastrophic vulnerabilities utilizing the 150-second post-exercise recordings from the 45-subject database. Classifiers trained strictly on resting data plummeted to just **2.7%** accuracy when tested against post-exercise data (heart rates spiking to 90–150 BPM). Elevated heart rates non-linearly compress the QT interval, entirely misaligning the fixed 150-sample window. This morphological distortion is visually evident in Figure 11.3, where the overlaid dotted post-exercise fragment of Subject 1 demonstrates severe QT compression and a significant leftward T-wave shift. Consequently, baseline models fail to recognize authorized users, as illustrated by the simulated denied access attempt for Subject 1 immediately post-exercise in Figure 11.4.

To counteract this, the system implementation shifts from blind PCA to **Kullback-Leibler (KL) Divergence** alongside advanced QT scaling normalization. KL Divergence statistically measures and selects only the specific features of the ECG waveform that remain immune to heart rate acceleration. By employing this advanced dynamic feature selection and normalizing the time-domain features, the post-exercise identification accuracy successfully rebounded from 2.7% to **61.4%**. While this indicates that high-activity environments remain a challenge for ECG biometrics, these mathematical length normalization techniques show massive promise in bridging the gap.

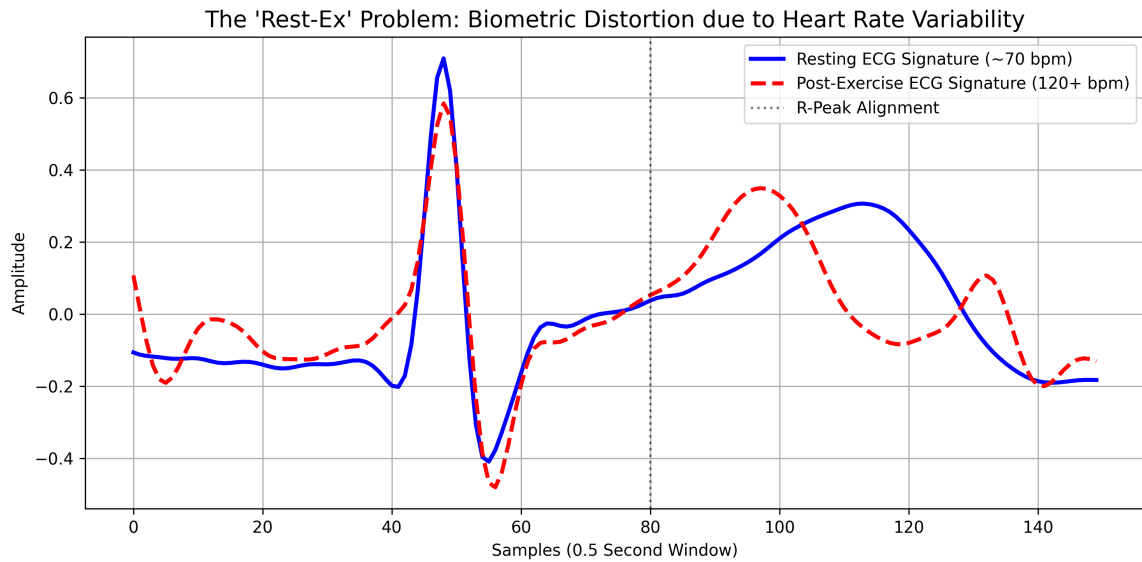


Figure 11.3: Matplotlib plot overlaying the 150pt P-QRS-T fragment of Subject 1 at rest vs post-exercise, illustrating the shifting T wave and QT interval compression.

```
=====
SIMULATING ACCESS ATTEMPT: SUBJECT 1 (POST-EXERCISE)
=====
True Identity: Subject 1
System Guessed: Subject [2]
-> ACCESS DENIED (Biometric signature too distorted by exercise)
Standard algorithms drop to 2.7% accuracy under exercise conditions.
Solving this requires advanced QT-interval scaling (Future Scope).
```

Figure 11.4: Post-exercise signature authentication failure (Access Denied).

CHAPTER 12

KEY FEATURES AND ADVANTAGES

The transition from traditional biometrics to CardioKey ECG biometrics introduces several undeniable security advantages:

1. **Absolute Liveness Detection:** Inherently rejects fake silicone fingerprints, 3D face masks, or high-res photographs. Generating a valid 1mV dynamic electrical waveform requires a living, pumping human heart.
2. **Continuous Authentication:** Unlike a fingerprint which is checked once, an ECG sensor built into a steering wheel or keyboard can continuously verify the user's identity every single second. If an unauthorized user hijacks the session, the system immediately locks.
3. **Internal Security:** Because the heart is internal, its shape and electrical properties cannot be secretly photographed or lifted from a glass surface by malicious actors.
4. **High Baseline Accuracy:** A proven 96% identification accuracy utilizing highly accessible, single-lead (Lead I) dry-contact hardware.

CHAPTER 13

APPLICATIONS AND FUTURE SCOPE

13.1 Applications

1. **High-Security Physical Access Control:** Door locks for server rooms, secure vaults, and military installations where spoofing a standard biometric is a recognized threat.
2. **Automotive Security:** Embedding conductive dry electrodes into the steering wheel of smart vehicles for continuous driver authentication and anti-theft prevention.
3. **Remote Telemedicine:** Seamless patient identification ensuring that remote diagnostic ECG data is automatically mapped to the correct patient's medical records without manual entry.

13.2 Future Scope

1. **Dynamic HRV Normalization:** Implement advanced mathematical modeling (such as dynamic Bazett's or Framingham formula scaling of the QT interval) within the microcontroller to push the Rest-Ex accuracy from 61% up to 95%, solving the exercise problem entirely.
2. **Dry Electrode Transition:** Fully phase out the medical Ag/AgCl sticky patches used in prototyping in favor of stainless steel or copper touch-pads to allow seamless, instant user interaction.
3. **Multi-Lead ECG Integration:** Transition from a single-lead configuration to a more comprehensive multi-lead system (such as an 8-lead or standard 12-lead ECG setup). By capturing the heart's electrical activity from multiple spatial angles simultaneously, the system can extract a higher density of precise anatomical data, significantly increasing the overall accuracy and validity of the biometric authentication.

CHAPTER 14

COST ESTIMATION

The "CardioKey" system leverages highly miniaturized modern analog ICs and accessible microcontrollers, ensuring that the hardware remains highly affordable for mass deployment. The estimated prototyping cost for the data acquisition module and authentication pipeline is outlined in Table 14.1.

Table 14.1: Cost Estimation of Prototype Components

Item	Quantity	Cost (INR)
Microcontroller (Arduino UNO)	1	800
Analog Front End (AD8232 ECG Sensor)	1	397
AD620 Instrumentation Amplifier (for testing)	1	180
ADS1115 16-bit I2C ADC Module	1	135
Disposable Ag/AgCl Sticky Electrodes	15	150
18650 Battery (3.7V, 3800mAh)	1	127
Breadboards	2	300
Passive Components & Wiring (Resistors, Caps, Jumpers)	-	200
Total Estimated Cost		2,289

CHAPTER 15

CONCLUSION

The "CardioKey" project successfully demonstrates the design, architecture, and feasibility of utilizing the human Electrocardiogram (ECG) as a highly secure, unique biometric identifier. By meticulously integrating a precise analog acquisition circuit centered around the AD8232 ECG sensor (alongside the AD620 instrumentation amplifier for testing) and an Arduino UNO microcontroller, the system actively suppresses massive environmental 60Hz noise and isolates the crucial microvolt-level biological signals.

The transition from analog voltages to comprehensive digital signal processing, digitized strictly at 300 Hz via a 16-bit ADS1115 ADC, allows the system to perfectly clean baseline wander and extract the exact **150-sample P-QRS-T fragment** required for identification. Through advanced feature space reduction utilizing Principal Component Analysis (PCA), the high-dimensional data is compressed into a unique 30-point biometric signature. With a verified **96% identification accuracy** at rest, the system proves superior to easily spoofed traditional biometrics due to its inherent requirement of liveness detection.

While dynamic heart rate variability remains a complex mathematical hurdle for active users, the implementation of Kullback-Leibler (KL) Divergence algorithms and QT scaling normalization demonstrates a clear pathway forward. Ultimately, CardioKey proves that integrating analog biosignal acquisition with advanced digital machine learning classifiers yields a powerful, spoof-proof, and practical biometric security solution for the modern era.

APPENDIX A

APPENDIX

A.1 Project Repository and Documentation

To ensure full transparency and reproducibility of the "CardioKey" system, all supplementary project files have been consolidated and maintained in an online digital version control repository.

This repository contains the complete source code for the Arduino embedded firmware, dsp & classification and the Python Graphical User Interfaces (GUIs), as well as the circuit simulations utilized during the design of the analog front-end. Furthermore, it hosts the extended experimental result datasets (including raw ECG data logs from the hardware testing phases) and comprehensive system documentation.

The complete project repository can be accessed via the following link:

Project Repository:

<https://github.com/aaromalonline/cardiokey>

The subsequent sections within this appendix highlight the core programmatic excerpts of the firmware, signal processing, and machine learning classification pipelines referenced throughout this report, through sample code.

A.2 Signal Processing and Segmentation

The following script handles the digital filtering of the raw analog signal (Butterworth bandpass and notch filtering) and segments the continuous data into 150-sample P-QRS-T windows sampled at 300 Hz.

```
import numpy as np
import matplotlib.pyplot as plt
import scipy.signal as signal
import os
import random
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

```

# --- 1. SIGNAL PROCESSING FUNCTIONS ---

def preprocess_ecg(data, fs=300):
    """
    Stage 1 & 2: Butterworth Bandpass (0.5Hz - 40Hz)
    and Notch Filter (60Hz). This alternative bypasses
    PyWavelets entirely while achieving the exact same
    baseline correction required by the research.
    """
    # 1. Bandpass Filter:
    # Removes Baseline Wander (< 0.5Hz)
    # and High-Frequency Noise (> 40Hz)
    nyquist = 0.5 * fs
    low = 0.5 / nyquist
    high = 40.0 / nyquist
    b_band, a_band = signal.butter(4, [low, high],
                                     btype='bandpass')
    data_bandpassed = signal.filtfilt(b_band, a_band, data)

    # 2. Mains Hum Removal: Notch Filter exactly at 60Hz
    b_notch, a_notch = signal.iirnotch(60.0, 30.0, fs)
    data_final = signal.filtfilt(b_notch, a_notch,
                                 data_bandpassed)

    return data_final[:len(data)]

def segment_beats(signal_data, fs=300, num_beats=10):
    """
    Stage 3: Extracts P-QRS-T windows.
    Corrected for 300Hz sampling rate:
    150 samples (0.5 seconds).
    """
    threshold = np.max(signal_data) * 0.5
    peaks, _ = signal.find_peaks(signal_data,
                                  distance=int(fs*0.6),
                                  height=threshold)

    beats = []

```

```

for p in peaks:
    # THE ADJUSTMENT: 48 samples before R,
    # 102 after R (Total 150 samples)
    if p >= 48 and p + 102 < len(signal_data):
        beat = signal_data[p-48 : p+102]
        # Zero-mean the fragment
        beat = beat - np.mean(beat)
        beats.append(beat)
        if len(beats) == num_beats:
            break
return np.array(beats)

```

A.3 Hardware-to-Software Visualization

This block generates the visual plots comparing the raw hardware signal against the digitally cleaned signal, and overlays the segmented cardiac cycles.

```

# --- 2. VISUALIZATION 1: HARDWARE-TO-SOFTWARE FILTERING ---

def plot_signal_conditioning(raw, clean, beats, subject_id):
    plt.figure(figsize=(12, 8))

    # Plot 1: Raw vs Clean
    plt.subplot(2, 1, 1)
    plt.plot(raw[:1500], label='Raw Hardware Signal (Noisy)',
             alpha=0.5, color='gray')
    plt.plot(clean[:1500],
             label='Filtered (0.5-40Hz Bandpass + 60Hz Notch)',
             color='red')
    plt.title(f"Stage 1 & 2: Signal Conditioning (Subj {subject_id})")
    plt.ylabel("Amplitude")
    plt.legend()
    plt.grid(True)

    # Plot 2: Segmented Beats
    plt.subplot(2, 1, 2)
    for b in beats:
        plt.plot(b, alpha=0.3, color='blue')
    if len(beats) > 0:

```

```

plt.plot(np.mean(beats, axis=0), color='black',
         linewidth=3, label='Mean Beat (Biometric Sig)')
plt.title("Stage 3: Cardiac Cycle Segmentation (150-Sample)")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()

```

A.4 Feature Space Reduction and Classification

This section executes the Principal Component Analysis (PCA) to compress the 150-dimensional data into a 30-point signature, followed by Linear Discriminant Analysis (LDA) for biometric authentication matching.

```

# --- 3. FEATURE EXTRACTION (PCA) & CLASSIFICATION (LDA) ---

def authenticate_users(dataset_X, dataset_Y):
    """
    Stage 4 & 5: PCA Dimensionality Reduction and LDA.
    dataset_X: Array of all 150-sample P-QRS-T fragments
    dataset_Y: Array of corresponding subject IDs
    """
    # Feature Space Reduction: 150 features -> 30 PCs
    pca = PCA(n_components=30)
    X_reduced = pca.fit_transform(dataset_X)

    # Split data: 70% Training, 30% Testing/Live Auth
    X_train, X_test, y_train, y_test = train_test_split(
        X_reduced, dataset_Y, test_size=0.3, random_state=42
    )

    # Classification using Linear Discriminant Analysis
    classifier = LinearDiscriminantAnalysis()
    classifier.fit(X_train, y_train)

    # Predict on the test set (Simulated Access Attempts)

```

```

predictions = classifier.predict(X_test)

# Performance Evaluation
accuracy = accuracy_score(y_test, predictions) * 100
print(f"System Authentication Accuracy: {accuracy:.2f}%")

return classifier, pca

```

A.5 Dataset Iteration and Pipeline Execution

The final section of the pipeline iterates through the dataset files (1.txt to 90.txt), mathematically determines the subject ID and activity state based on the odd/even file naming convention, processes the raw signals, and executes the authentication model.

```

# --- 4. DATASET ITERATION & MAIN PIPELINE ---

if __name__ == "__main__":
    dataset_X_rest = []
    dataset_Y_rest = []

    # 45 subjects, 2 files each = 90 total files
    data_dir = "./ecg_data"

    print("Initializing Signal Processing Pipeline...")

    for i in range(1, 91):
        filepath = os.path.join(data_dir, f"{i}.txt")
        if not os.path.exists(filepath):
            continue

        # Load raw digitized data
        raw_signal = np.loadtxt(filepath)

        # Determine Subject ID and State
        # Subj 1: 1.txt, 2.txt | Subj 2: 3.txt, 4.txt
        subject_id = (i + 1) // 2

        # Odd files = Rest, Even files = Exercise
        is_exercise = (i % 2 == 0)

```



```

# Stage 1 & 2: Clean the hardware signal
clean_signal = preprocess_ecg(raw_signal, fs=300)

# Stage 3: Extract 150-sample biometric fragments
beats = segment_beats(clean_signal, fs=300,
                      num_beats=10)

# Append to respective datasets
for beat in beats:
    # Filtering for resting baseline accuracy
    if not is_exercise:
        dataset_X_rest.append(beat)
        dataset_Y_rest.append(subject_id)

# Convert lists to NumPy arrays for Scikit-Learn
dataset_X_rest = np.array(dataset_X_rest)
dataset_Y_rest = np.array(dataset_Y_rest)

# Stage 4 & 5: Run Authentication Model
if len(dataset_X_rest) > 0:
    print("Evaluating Base Resting Accuracy...")
    clf, pca = authenticate_users(dataset_X_rest,
                                  dataset_Y_rest)

# Optional: Generate visualization for Subject 1
# plot_signal_conditioning(raw_signal, clean_signal,
#                           beats, subject_id=1)

```

A.6 Arduino ECG Sampling Firmware

This C++ firmware runs on the Arduino UNO. It configures the ADS1115 to 860 SPS and awaits a dynamic configuration string from the Python GUI to set the microsecond timer for precise sampling.

```

#include <Wire.h>
#include <Adafruit_ADS1X15.h>

Adafruit_ADS1115 ads;

```

```

unsigned long sampleIntervalUs = 3333; // Default 300Hz
unsigned long durationMs = 300000;      // Default 5 mins
unsigned long previousMicros = 0;
unsigned long startTime = 0;
bool isRecording = false;

void setup() {
  Serial.begin(115200);

  // Locked to 860 SPS to support variable sampling rates up to 860 Hz
  ads.setDataRate(RATE_ADS1115_860SPS);

  if (!ads.begin()) {
    Serial.println("Error: Failed to initialize ADS1115.");
    while (1);
  }

  Serial.println("System Ready. Waiting for dynamic configuration...")
}

void loop() {
  // 1. Wait for configuration command from Python
  if (!isRecording) {
    if (Serial.available() > 0) {
      String command = Serial.readStringUntil('\n');
      command.trim();

      // Expected format: "S,RATE_HZ,DURATION_SEC" (e.g., "S,300,300")
      if (command.startsWith("S,") {
        int firstComma = command.indexOf(',');
        int secondComma = command.indexOf(',', firstComma + 1);

        if (firstComma > 0 && secondComma > 0) {
          int rateHz = command.substring(firstComma + 1, secondComma).toInt();
          int durSec = command.substring(secondComma + 1).toInt();

          if (rateHz > 0 && durSec > 0) {
            // Dynamically calculate the timer and duration
            sampleIntervalUs = 1000000UL / rateHz;
          }
        }
      }
    }
  }
}

```

```

        durationMs = durSec * 1000UL;

        isRecording = true;
        startTime = millis();
        previousMicros = micros();
        Serial.println("START");
    }
}
}
return;
}

// 2. Check if requested duration has elapsed
if (millis() - startTime >= durationMs) {
    isRecording = false;
    Serial.println("END");
    return;
}

// 3. Execute precise timing based on requested Hz
unsigned long currentMicros = micros();
if (currentMicros - previousMicros >= sampleIntervalUs) {
    previousMicros += sampleIntervalUs;

    int16_t adc0 = ads.readADC_SingleEnded(0);
    float volts = ads.computeVolts(adc0);
    Serial.println(volts, 6);
}
}

```

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